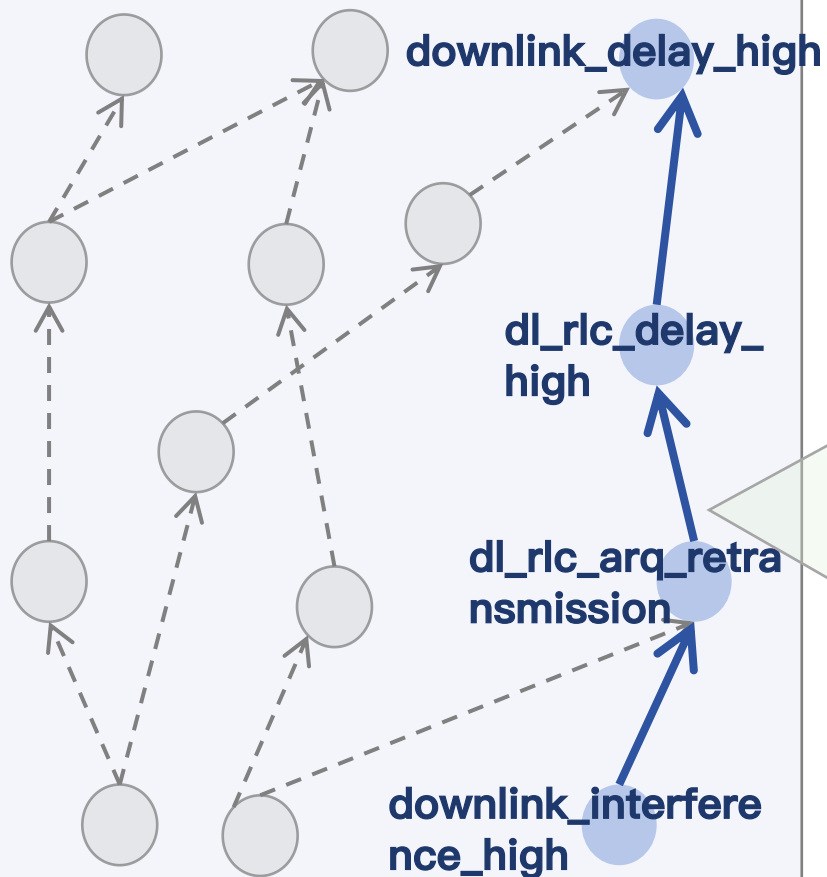


Global Causal Graph



Expert-Defined Rules

Downlink Interference

- DL_RLC_AVG_DELAY_SUM $\geq$ 200 ms
- DL_RBLER $>$ 0.1 or UL_RBLER $>$ 0.1
- SERVING_CELL_SSB_RSRQ $<$ -15
- ...

Uplink Weak Coverage

- CELL_DL_PRB_RATIO $>$ 0.7
- ...
- ...
- ...

Edge-Level Causal Constraint

Mechanism

Frequent ARQ retransmissions and reordering keep SDUs in RLC buffer longer, directly inflating RLC queuing and reordering delay.

Key Signals

RLC ARQ retransmission increase:
DL_RLC_ARQ_RETRANSMISSION_COUNT

Delay indicators increase:
DL_RLC_AVG_DELAY_SUM

Low-delay buckets:
DL_RLC_DELAY_IN_0_5MS_COUNT



Fault-Atom

-----> Causal Edge



Root Cause Category



Network KPI